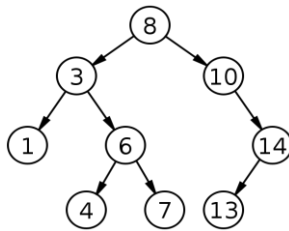

COMP232 – Data Structures & Problem Solving
Fall 2024

Homework #4
Binary Trees
[45 points]

Note: Answer #1 and #2 here. When you have completed all parts of this homework, save this document as PDF by creating a folder named “noncode-answers” in your git repository for this homework assignment, then push your code to GitHub. Complete the rest of the questions in Eclipse and push to GitHub.

1. Answer the questions below with respect to the following binary tree using the definitions from the text:



- a. What value is at the root of the tree? 8
 - b. What values are in leaves of the tree? 1, 4, 7, 13
 - c. What values are in internal nodes of the tree? 8, 3, 10, 6, 14
 - d. What values are descendants of 3? 1, 6, 4, 7
 - e. What values are ancestors of 7? 6, 3, 8
 - f. What is the depth of node 6? 2
 - g. What is the height of the tree? 3
 - h. What is the path length from 8 to 4? 3
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- i. List the node values in the order they would be visited by:
 - i. An inorder traversal. $1 \rightarrow 3 \rightarrow 4 \rightarrow 6 \rightarrow 7 \rightarrow 8 \rightarrow 10 \rightarrow 13 \rightarrow 14$
 - ii. A preorder traversal. $8 \rightarrow 3 \rightarrow 1 \rightarrow 6 \rightarrow 4 \rightarrow 7 \rightarrow 10 \rightarrow 14 \rightarrow 13$
 - iii. A postorder traversal. $1 \rightarrow 4 \rightarrow 7 \rightarrow 6 \rightarrow 3 \rightarrow 13 \rightarrow 14 \rightarrow 10 \rightarrow 8$
 - iv. A levelorder traversal. $8 \rightarrow 3 \rightarrow 10 \rightarrow 1 \rightarrow 6 \rightarrow 14 \rightarrow 4 \rightarrow 7 \rightarrow 13$

2. Assuming the variable `node` refers to a `BTNode` and that the tree is always sufficiently large, give a statement that identifies the `node`'s:

- a. leftmost grand child.
 - This is the left child node of the left child node of a node. To access the leftmost grandchild, we have to access the left node of a given node, then access the left node of that child. (node.left.left)
- b. great grand parent.
 - This is the parent node of the parent node of the parent node of a node. (node.parent.parent.parent)
- c. left child of right sibling, assuming node is a left child.
 - If a node is a left child, so its right sibling is node.parent.right. And the left child of the right sibling is node.parent.right.left .

3. Complete the implementation of the `MinKeyFinder` class in the `hw04` package. The `No3Tests` class contains tests that you can use to check your implementation of this class.

4. Implement the `main` method in the `MinKeyFinder` class in the `hw04` package so that it creates a tree with at least 10 nodes and uses a preorder traversal with your visitor to print the key associated with the minimum value in the tree.

5. Complete the implementation of the `visitInOrder` method in the `CS232LinkedBinaryTree` class in the `hw04` package. The `No5Tests` class contains tests that you can use to check your implementation of this method.

6. Complete the implementation of the three-argument constructor for the `CS232LinkedBinaryTree` class in your `hw04` package. The `No6Tests` class contains tests that you can use to check your implementations of this constructor.

7. Complete the implementation of the `contains` method in the `CS232LinkedBinaryTree` class in the `hw04` package. The `No7Tests` class contains tests that you can use to check your implementations of this method. For full credit, implement `contains` so that it does not rely on the `get` or `getNodeFromSubtree` methods. Instead, add a helper method `subtreeContains` that returns a boolean value. This will be similar to the `getNodeFromSubtree` method. Then call `subtreeContains` from `contains`. Yes, this introduces repeated and unnecessary code, but the point here is not producing the most effective class, but for you to practice writing methods like this.

8. Complete the implementation of the `add` method in the `CS232LinkedBinaryTree` class in the `hw04` package. The `No8Tests` class contains tests that you can use to check your implementations of this method.

9. Complete the implementation of the `countLeafNodes` method in the `CS232LinkedBinaryTree` class in the `hw04` package. The `No9Tests` class contains tests that you can use to check your implementations of this method.